Definitions in Analysis I by Terence Tao

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Contents

Remark. This document follows the third edition of *Analysis I*.

1 Introduction

No definitions found in this chapter.

2 Starting at the beginning: the natural numbers

2.1 The Peano axioms

Axiom 2.1. 0 is a natural number.

Axiom 2.2. If n is a natural number, then $n{++}$ is also a natural number.

Definition 2.1.3. We define 1 to be the number 0++, 2 to be the number (0++)++, 3 to be the number ((0++)++)++, etc. In other words, $1 \coloneqq 0{++}$, $2 \coloneqq 1{++}$, $3 \coloneqq 2{++}$, etc.

Axiom 2.3. 0 is not the successor of any natural number, i.e., we have $n{++} \neq 0$ for every natural number n.

Axiom 2.4. Different natural numbers must have different successors; i.e. if n, m are natural numbers and $n \neq m$, then $n{++} = m{++}$. Equivalently, if $n{++} = m{++}$, then we must have $n = m$.

Axiom 2.5 (Principle of mathematical induction)**.** Let $P(n)$ be any property pertaining to a natural number n. Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n{++})$ is also true. Then $P(n)$ is true for every natural number n.

2.2 Addition

Definition 2.2.1 (Addition of natural numbers)**.** Let m be a natural number. To add 0 to m, we define $0{+}m \coloneqq m$. Now, suppose inductively that we have defined how to add n to m. Then, we can add $n{++}$ to m by defining $(n{++}) + m \coloneqq (n + m){++}$.

Lemma 2.2.2. For any natural number n, $n + 0 = n$.

Lemma 2.2.3. For any natural numbers n and m, $n + (m{++}) = (n + m){++}$.

Proposition 2.2.4 (Addition is commutative)**.** For any natural numbers n and m, $n + m = m + n$.

Proposition 2.2.5 (Addition is associative)**.** For any natural numbers a, b, c, we have $(a + b) + c = a + (b + c)$.

Proposition 2.2.6 (Cancellation law)**.** Let a, b, c be natural numbers such that $a + b = a + c$. Then we have $b = c$.

Definition 2.2.7 (Positive natural numbers)**.** A natural number b is said to be *positive* iff it is not equal to 0.

Proposition 2.2.8. If a is positive and b is a natural number, then $a+b$ is positive.

Corollary 2.2.9. If a and b are natural numbers such that $a + b = 0$, then $a = 0$ and $b = 0$.

Lemma 2.2.10. Let a be a positive number. Then there exists exactly one natural number b such that $b{+}{+} = a$.

Definition 2.2.11 (Ordering of the natural numbers)**.** Let n and m be natural numbers. We say that n is greater than or equal to m, and write $n \geq m$ or $m \leq n$, iff we have $n = m + a$ for some natural number a. We say that n is *strictly greater than* m, and write $n > m$ or $m < n$, iff $n \geq m$ and $n \neq m$.

Proposition 2.2.12 (Basic properties of order for natural numbers)**.** Let a, b, c be natural numbers. Then

(a) *Reflexivity:* $a \geq a$.

(b) *Transitivity: If* $a \geq b$ *and* $b \geq c$, *then* $a \geq c$.

(c) *Anti-symmetry: If* $a \geq b$ *and* $b \geq a$, *then* $a = b$.

(d) *Preservation of order:* $a \geq b$ *if and only if* $a + c \geq b + c$.

(e) $a < b$ *if and only if* $a{+}{+} \leq b$.

(f) $a < b$ *if and only if* $b = a + d$ *for some positive number* d.

Proposition 2.2.13 (Trichotomy of order for natural number)**.** Let a and b be natural numbers. Then exactly one of the following statements is true: $a < b$, $a = b$, or $a > b$.

Proposition 2.2.14 (Strong principle of induction)**.** Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m. Suppose that for each $m \geq m_0$, we have the following implication: if $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. Then we can conclude that $P(m)$ is true for all natural numbers $m \geq m_0$.

2.3 Multiplication

Definition 2.3.1 (Multiplication of natural numbers)**.** Let m be a natural number. To multiply 0 to m, we define $0 \times m := 0$. Now, suppose inductively that we have defined how to multiply n to m. Then, we can multiply n++ to m by defining $(n{+}{+}) \times m := (n \times m) + m$.

Lemma 2.3.2 (Multiplication is commutative)**.** Let n, m be natural numbers. Then $n \times m = m \times n$.

Lemma 2.3.3 (Positive natural numbers have no zero divisors)**.** Let n, m be natural numbers. Then $n \times m = 0$ if and only if at least one of n, m is equal to 0. In particular, if n and m are both positive, then nm is also positive.

Proposition 2.3.4 (Distributive law)**.** For any natural numbers a, b, c, we have $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

Proposition 2.3.5 (Multiplication is associative)**.** For any natural numbers a, b, c, we have $(a \times b) \times c = a \times (b \times c)$

Proposition 2.3.6 (Multiplication preserves order)**.** If a, b are natural numbers such that $a < b$, and c is positive, then $ac < bc$.

Corollary 2.3.7 (Cancellation law)**.** Let a, b, c be natural numbers such that $ac = bc$ and c is nonzero. Then $a = b$.

Proposition 2.3.9 (Euclidean algorithm)**.** Let n be a natural number, and let q be a positive number. Then there exist natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$.

Definition 2.3.11 (Exponentiation for natural numbers)**.** Let m be a natural number. To raise m to the power 0, we define $m^0 := 1$; in particular, we define $0^0 := 1$. Now suppose recursively that m^n has been defined for some natural number n, then we define $m^{n++} := m^n \times m$.

3 Set theory

3.1 Fundamentals

Axiom 3.1 (Sets are objects)**.** If A is a set, then A is also an object. In particular, given two sets A and B, it is meaningful to ask whether A is also an element of B.

Definition 3.1.4 (Equality of sets)**.** Two sets A and B are *equal*, $A = B$, iff every element of A is an element of B and vice versa. To put it another way, $A = B$ if and only if every element x of A belongs also to B, and every element y of B belongs also to A.

Axiom 3.2 (Empty set)**.** There exists a set $\varnothing$, known as the *empty set*, which contains no elements, i.e., for every object x we have $x \notin \varnothing$.

Lemma 3.1.6 (Single choice)**.** Let A be a non-empty set. Then there exists an object x such that $x \in A$.

Axiom 3.3 (Singleton sets and pair sets)**.** If a is an object, then there exists a set $\{a\}$ whose only element is a, i.e., for every object y, we have $y \in \{a\}$ if and only if $y = a$; we refer to $\{a\}$ as the singleton set whose element is a. Furthermore, if a and b are objects, then there exists a set $\{a, b\}$ whose only elements are a and b; i.e., for every object y, we have $y \in \{a, b\}$ if and only if $y = a$ or $y = b$; we refer to this set as the pair set formed by a and b.

Axiom 3.4 (Pairwise union)**.** Given any two sets A, B, there exists a set $A \cup B$, called the *union* $A \cup B$ of A and B, whose elements consist of all the elements which belong to A, or B or both. In other words, for any object x,

$$x \in A \cup B \iff (x \in A \text{ or } x \in B).$$

Lemma 3.1.13. If a and b are objects, then $\{a, b\} = \{a\} \cup \{b\}$. If A, B, C are sets, then the union operation is commutative (i.e., $A \cup B = B \cup A$) and associative (i.e., $(A \cup B) \cup C = A \cup (B \cup C)$). Also, we have $A \cup A = A \cup \varnothing = \varnothing \cup A = A$.

Definition 3.1.15 (Subsets)**.** Let A, B be sets. We say that A is a *subset* of B, denoted $A \subseteq B$, iff every element of A is also an element of B, i.e.

$$\text{For any object } x, \quad x \in a \implies x \in B.$$

We say that A is a *proper subset* of B, denoted $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Proposition 3.1.18 (Sets are partially ordered by set inclusion)**.** Let A, B, C be sets. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. If $A \subseteq B$ and $B \subseteq A$, then $A = B$. Finally, if $A \subsetneq B$ and $B \subsetneq C$, then $A \subsetneq C$.

Axiom 3.5 (Axiom of specification/axiom of separation)**.** Let A be a set, and for each $x \in A$, let $P(x)$ be a property pertaining to x (i.e., $P(x)$ is either a true statement or a false statement). Then there exists a set, called $\{x \in A : P(x) \text{ is true}\}$ (or simply $\{x \in A : P(x)\}$ for short), whose elements are precisely the elements x in A for which $P(x)$ is true. In other words, for any object y,

$$y \in \{x \in A : P(x) \text{ is true}\} \iff (y \in A \text{ and } P(y) \text{ is true}).$$

Definition 3.1.23 (Intersections)**.** The *intersection* $S_1 \cap S_2$ of two sets is defined to be the set

$$S_1 \cap S_2 \coloneqq \{x \in S_1 : x \in S_2\}.$$

Definition 3.1.27 (Difference set)**.** Given two sets A and B, we define the set $A - B$ or $A \setminus B$ to be the set A with any elements of B removed:

$$A \setminus B \coloneqq \{x \in A : x \notin B\}.$$

Proposition 3.1.28 (Sets form a boolean algebra)**.** Let A, B, C be sets, and let X be a set containing A, B, C as subsets.

(a) (Minimal element) We have $A \cup \varnothing = A$ and $A \cap \varnothing = \varnothing$.

(b) (Maximal element) We have $A \cup X = X$ and $A \cap X = A$.

(c) (Identitiy) We have $A \cap A = A$ and $A \cup A = A$.

(d) (Commutativity) We have $A \cup B = B \cup A$ and $A \cap B = B \cap A$.

(e) (Associativity) We have $(A \cup B) \cup C = A \cup (B \cup C)$and $(A \cap B) \cap C = A \cap (B \cap C)$.

(f) (Distributivity) We have $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

(g) (Partition) We have $A \cup (X \setminus A) = X$ and $A \cap (X \setminus A) = \varnothing$.

(h) (De Morgan Laws) We have $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ and $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$.

Axiom 3.6 (Replacement)**.** Let A be a set. For any object $x \in A$ and any object y, suppose we have a statement $P(x, y)$ pertaining to x and y, such that for each $x \in A$ there is at most one y for which $P(x, y)$ is true. Then there exists a set $\{y : P(x, y) \text{ is true for some } x \in A\}$, such that for any object z,

$$z \in \{y : P(x, y) \text{ is true for some } x \in A\} \iff P(x, z) \text{ is true for some } x \in A.$$

Axiom 3.7 (Infinity)**.** There exists a set $\mathbb{N}$, whose elements are called natural numbers, as well as an object 0 in $\mathbb{N}$, and an object $n{+}{+}$ assigned to every natural number $n \in \mathbb{N}$, such that the Peano axioms (Axioms 2.1 - 2.5) hold.

3.2 Russell's paradox

Axiom 3.8 (Universal specification)**.** (*Dangerous!*) Suppose for every object x we have a property $P(x)$ pertaining to x (so that for every x, $P(x)$ is either a true statement or a false statement). Then there exists a set $\{x : P(x) \text{ is true}\}$ such that for every object y,

$$y \in \{x : P(x) \text{ is true}\} \iff P(y) \text{ is true}.$$

Axiom 3.9 (Regularity)**.** If A is a non-empty set, then there is at least one element x of A which is either not a set, or is disjoint from A.

3.3 Functions

Definition 3.3.1 (Functions)**.** Let X, Y be sets, and let $P(x, y)$ be a property pertaining to an object $x \in X$ and an object $y \in Y$, such that for every $x \in X$, there is exactly one $y \in Y$ for which $P(x, y)$ is true (this is sometimes known as the *vertical line test*). Then we define the *function $f : X \to Y$ defined by P on the domain X and range Y* to be the object which, given any input $x \in X$, assigns an output $f(x) \in Y$, defined to be the unique object $f(x)$ for which $P(x, f(x))$ is true. Thus, for any $x \in X$ and $y \in Y$,

$$y = f(x) \iff P(x, y) \text{ is true}.$$

Definition 3.3.7 (Equality of functions)**.** Two functions $f : X \to Y$, $g : X \to Y$ with the same domain and range are said to be equal, $f = g$, if and only if $f(x) = g(x)$

for all $x \in X$. (If $f(x)$ and $g(x)$ agree for some values of x, but not others, then we do not consider f and g to be equal.)

Definition 3.3.10 (Composition)**.** Let $f : X \to Y$ and $g : Y \to Z$ be two functions, such that the range of f is the same set as the domain of g. We then define the composition $g \circ f : X \to Z$ of the two functions g and f to be the function defined explicitly by the formula

$$(g \circ f)(x) := g(f(x)).$$

If the range of f does not match the domain of g, we leave the composition $g \circ f$ undefined.

Lemma 3.1.12 (Composition is associative)**.** Let $f : Z \to W$, $g : Y \to Z$, and $h : X \to Y$ be functions. Then $f \circ (g \circ h) = (f \circ g) \circ h$.

Definition 3.3.14 (One-to-one function)**.** A function f is *one-to-one* (or *injective*) if different elements map to different elements:

$$x \neq x' \implies f(x) \neq f(x').$$

Equivalently, a function is one-to-one if

$$f(x) = f(x') \implies x = x'.$$

Definition 3.3.17 (Onto functions)**.** A function f is *onto* (or *surjective*) if $f(X) = Y$, i.e., every element in Y comes from applying f to some element in X:

For every $y \in Y$, there exists $x \in X$ such that $f(x) = y$.

Definition 3.3.20 (Bijective functions)**.** Functions $f : X \to Y$ which are both one-to-one and onto are also called *bijective* or *invertible*.

3.4 Images and inverse images

Definition 3.4.1 (Images of sets)**.** If $f : X \to Y$ is a function from X to Y, and S is a set in X, we define $f(S)$ to be the set

$$f(S) := \{f(x) : x \in S\};$$

this set is a subset of Y, and is sometimes called the *image* of S under the map f. In other words,

$$x \in S \implies f(x) \in f(S),$$

and

$$y \in f(S) \iff y = f(x) \text{ for some } x \in S.$$

Definition 3.4.4 (Inverse images)**.** If U is a subset of Y, we define the set $f^{-1}(U)$ to be the set

$$f^{-1}(U) := \{x \in X : f(x) \in U\}.$$

In other words, $f^{-1}(U)$ consists of all the elements of X which map into U:

$$f(x) \in U \iff x \in f^{-1}(U).$$

We call $f^{-1}(U)$ the inverse image of U.

Axiom 3.10 (Power set axiom)**.** Let X and Y be sets. Then there exists a set, denoted Y^X, which consists of all the functions from X to Y, thus

$$f \in Y^X \iff (f \text{ is a function with domain } X \text{ and range } Y).$$

Lemma 3.4.9. Let X be a set. Then the set

$$\{Y : Y \text{ is a subset of } X\}$$

is a set. This set is known as the *power set* of X and is denoted 2^X.

Axiom 3.11 (Union)**.** Let A be a set, all of whose elements are themselves sets. Then there exists a set $\bigcup A$ whose elements are precisely those objects which are elements of the elements of A, thus for all objects x

$$x \in \bigcup A \iff (x \in S \text{ for some } S \in A).$$

If one has some set I, and for every element $\alpha \in I$ we have some set A_α, then we can form the union set $\bigcup_{\alpha \in I} A_\alpha$ by defining

$$\bigcup_{\alpha \in I} A_\alpha := \bigcup \{A_\alpha : \alpha \in I\}.$$

More generally, for any objects y,

$$y \in \bigcup_{\alpha \in I} A_\alpha \iff (y \in A_\alpha \text{ for some } \alpha \in I). \tag{3.2}$$

We often refer to I as an *index set*, and the elements α of this index set as *labels*; the sets A_α are then called a *family of sets*, and are indexed by the labels $\alpha \in I$.

Unlabeled Definition (Intersection)**.** Given any non-empty set I, and given an assignment of a set A_α to each $\alpha \in I$, we can define the intersection $\bigcap_{\alpha \in I} A_\alpha$ by first choosing some element β of I (which we can do since I is non-empty), and setting

$$\bigcap_{\alpha \in I} A_\alpha := \{x \in A_\beta : x \in A_\alpha \text{ for all } \alpha \in I\}. \tag{3.3}$$

Then, for any object y,

$$y \in \bigcap_{\alpha \in I} A_\alpha \iff (y \in A_\alpha \text{ for all } \alpha \in I). \tag{3.4}$$

3.5 Cartesian products

Definition 3.5.1 (Ordered pair)**.** If x and y are any objects (possibly equal), we define the ordered pair (x, y) to be a new object, consisting of x as its first component and y as its second component. Two ordered pairs (x, y) and (x', y') are considered equal if and only if both their components match, i.e.

$$(x, y) = (x', y') \iff (x = x' \text{ and } y = y'). \tag{3.5}$$

Definition 3.5.4 (Cartesian product)**.** If X and Y are sets, then we define the Cartesian product $X \times Y$ to be the collection of ordered pairs, whose first component lies in X and second component lies in Y, thus

$$X \times Y = \{(x, y) : x \in X, y \in Y\}$$

or equivalently

$$a \in (X \times Y) \iff (a = (x, y) \text{ for some } x \in X \text{ and } y \in Y).$$

Definition 3.5.7 (Ordered n-tuple and n-fold Cartesian product)**.** Let n be a natural number. An ordered n-tuple $(x_i)_{1 \leq i \leq n}$ (also denoted $(x_1, \ldots, x_n)$) is a collection of objects x_i, one for every natural number i between 1 and n; we refer to xi as the i^{th} *component* of the ordered n-tuple.

Two ordered n-tuples $(x_i)_{1 \leq i \leq n}$ and $(y_i)_{1 \leq i \leq n}$ are said to be equal iff $x_i = y_i$ for all $1 \leq i \leq n$.

If $(X_i)_{1\leq i\leq n}$ is an ordered n-tuple of sets, we define their *Cartesian product* $\prod_{1\leq i\leq n} X_i$ (also denoted $\prod_{i=1}^n X_i$ or $X_1 \times \ldots \times X_n$) by

$$\prod_{1\leq i\leq n} X_i \coloneqq \{(x_i)_{1\leq i\leq n} : x_i \in X_i \text{ for all } 1 \leq i \leq n\}.$$

If n is a natural number, we often write X^n as shorthand for the n-fold Cartesian product $X^n \coloneqq \prod_{1\leq i\leq n} X$.

Lemma 3.5.12 (Finite choice)**.** Let $n \geq 1$ be a natural number, and for each natural number $1 \leq i \leq n$, let X_i be a non-empty set. Then there exists an n-tuple $(x_i)_{1\leq i\leq n}$ such that $x_i \in X_i$ for all $1 \leq i \leq n$. In other words, if each X_i is non-empty, then the set $\prod_{1\leq i\leq n} X_i$ is also non-empty.

3.6 Cardinality of sets

Definition 3.6.1 (Equal cardinality)**.** We say that two sets X and Y have *equal cardinality* iff there exists a bijection $f : X \to Y$ from X to Y.

Proposition 3.6.4. Let X, Y, and Z be sets. Then X has equal cardinality with X. If X has equal cardinality with Y, then Y has equal cardinality with X. If X has equal cardinality with Y and Y has equal cardinality with Z, then X has equal cardinality with Z.

Definition 3.6.5. Let n be a natural number. A set X is said to have *cardinality* n, iff it has equal cardinality with $\{i \in \mathbb{N} : 1 \leq i \leq n\}$. We also say that X *has* n *elements* iff it has cardinality n.

Proposition 3.6.8 (Uniqueness of cardinality)**.** Let X be a set with some cardinality n. Then X cannot have any other cardinality, i.e., X cannot have cardinality m for any $m \neq n$.

Lemma 3.6.9. Suppose that $n \geq 1$, and X has cardinality n. Then X is non-empty, and if x is any element of X, then the set $X - \{x\}$ (i.e., X with the element x removed) has cardinality $n - 1$.

Definition 3.6.10 (Finite sets)**.** A set is *finite* iff it has cardinality N for some natural number n; otherwise, the set is called *infinite.* If X is a finite set, we use $\#(X)$ to denote the cardinality of X.

Theorem 3.6.12. The set of natural numbers $\mathbb{N}$ is infinite.

Proposition 3.6.14 (Cardinal arithmetic)**.**

(a) Let X be a finite set, and let x be an object which is not an element of X. Then $X \cup \{x\}$ is finite and $\#(X \cup \{x\}) = \#(X) + 1$.

(b) Let X and Y be finite sets. Then $X \cup Y$ is finite and $\#(X \cup Y) \leq \#(X) + \#(Y)$. If in addition X and Y are disjoint (i.e., $X \cap Y = \varnothing$), then $\#(X \cup Y) = \#(X) + \#(Y)$.

(c) Let X be a finite set, and let Y be a subset of X. Then Y is finite, and $\#(Y) \leq \#(X)$. If in addition $Y \neq X$ (i.e., Y is a proper subset of X), then we have $\#(Y) < \#(X)$.

(d) If X is a finite set, and $f : X \to Y$ is a function, then $f(X)$ is a finite set with $\#(f(X)) \leq \#(X)$. If in addition f is one-to-one, then $\#(f(X)) = \#(X)$.

(e) Let X and Y be finite sets. Then Cartesian product $X \times Y$ is finite and $\#(X \times Y) = \#(X) \times \#(Y)$.

(f) Let X and Y be finite sets. Then the set Y^X (defined in Axiom 3.10) is finite and $\#(Y^X) = \#(Y)^{\#(X)}$.

4 Integers and rationals

4.1 The integers

Definition 4.1.1 (Integers)**.** An *integer* is an expression of the form a — b, where a and b are natural numbers. Two integers are considered to be equal, a — $b = c$ — d, if and only if $a + d = c + b$. We let $\mathbb{Z}$ denote the set of all integers.

Definition 4.1.2. The sum of two integers, $(a \text{ — } b) + (c \text{ — } d)$, is defined by the formula

$$(a \text{ — } b) + (c \text{ — } d) := (a + c) \text{ — } (b + d).$$

The product of two integers $(a \text{ — } b) \times (c \text{ — } d)$, is defined by

$$(a \text{ — } b) \times (c \text{ — } d) := (ac + bd) \text{ — } (ad + bc).$$

Definition 4.1.4 (Negation of integers)**.** If $(a \text{ — } b)$ is an integer, we define the negation $-(a \text{ — } b)$ to be the integer $(b \text{ — } a)$. In particular, if $n = n \text{ — } 0$ is a positive natural number, we can define its negation $-n = 0 \text{ — } n$.

Lemma 4.1.5 (Trichotomy of integers)**.** Let x be an integer. Then exactly one of the following three statements is true: (a) x is zero; (b) x is equal to a positive natural number n; or (c) x is the negation $-n$ of a positive natural number n.

Unlabeled Definition. If n is a positive natural number, we call $-n$ a *negative integer*

Proposition 4.1.6 (Laws of algebra for integers)**.** Let x, y, z be integers. Then we have

$$\begin{aligned} x + y &= y + x \\ (x + y) + z &= x + (y + z) \\ x + 0 = 0 + x &= x \\ x + (-x) = (-x) + x &= 0 \\ xy &= yx \\ (xy)z &= x(yz) \\ x1 = 1x &= x \\ x(y + z) &= xy + xz \\ (y + z)x &= yx + zx. \end{aligned}$$

Remark. The above set of nine identities asserts that the integers form a *commutative ring.* If one deleted the identity $xy = yx$, then they would only assert that the integers form a *ring.*

Unlabeled Definition (Subtraction)**.** Define the operation of *subtraction* $x - y$ of two integers by the formula

$$x - y \coloneqq x + (-y).$$

Proposition 4.1.8 (Integers have no zero divisors)**.** Let a and b be integers such that $ab = 0$. Then either $a = 0$ or $b = 0$ (or both).

Corollary 4.1.9 (Cancellation law for integers)**.** If a, b, c are integers such that $ac = bc$ and c is non-zero, then $a = b$.

Definition 4.1.10 (Ordering of the integers)**.** Let n and m be integers. We say that n is *greater than or equal to* m, and write $n \geq m$ or $m \leq n$, iff we have $n = m + a$ for some natural number a. We say that n is *strictly greater than* m, and write $n > m$ or $m < n$, iff $n \geq m$ and $n \neq m$.

Lemma 4.1.11 (Properties of order)**.** Let a, b, c be integers.

(a) $a > b$ if and only if $a - b$ is a positive natural number.

(b) (Addition preserves order) If $a > b$, then $a + c > b + c$.

(c) (Positive multiplication preserves order) If $a > b$ and c is positive, then $ac > bc$.

(d) (Negation reverses order) If $a > b$, then $-a < -b$.

(e) (Order is transitive) If $a > b$ and $b.c$, then $a > c$.

(f) (Order trichotomy) Exactly one of the statements $a > b$, $a < b$, or $a = b$ is true.

4.2 The rationals

Definition 4.2.1. A *rational number* is an expression of the form $a//b$, where a and b are integers and b is non-zero; $a//0$ is not considered to be a rational number. Two rational numbers are considered to be equal, $a//b = c//d$, if and only if $ad = cb$. The set of all rational numbers is denoted $\mathbb{Q}$.

Definition 4.2.2. If $a//b$ and $c//d$ are rational numbers, we define their sum

$$(a//b) + (c//d) \coloneqq (ad = b)//(bd),$$

their product

$$(a//b) \times (c//d) \coloneqq (ac)//(bd),$$

and the negation

$$-(a//b) \coloneqq (-a)//b.$$

Unlabeled Definition (Reciprocal)**.** If $x = a//b$ is a non-zero rational (so that $a, b \neq 0$) then we define the *reciprocal* x^{-1} of x to be the rational number $x^{-1} \coloneqq b//a$. We leave the reciprocal of 0 undefined.

Proposition 4.2.4 (Laws of algebra for rationals)**.** Let x, y, z be rationals. Then the following laws of algebra hold:

$$\begin{aligned} x + y &= y + x \\ (x + y) + z &= x + (y + z) \\ x + 0 = 0 + x &= x \\ x + (-x) = (-x) + x &= 0 \\ xy &= yx \\ (xy)z &= x(yz) \\ x1 = 1x &= x \\ x(y + z) &= xy + xz \\ (y + z)x &= yx + zx. \end{aligned}$$

If x is non-zero, we also have

$$xx^{-1} = x^{-1}x = 1.$$

Remark. The above set of ten identities asserts that the rationals form a *field.*

Unlabeled Definition (Quotient)**.** The *quotient* x/y of two ratonal numbers x and y, provided that y is non-zero, is defined by the formula

$$x/y = x \times y^{-1}.$$

Definition 4.2.6. A rational number x is said to be *positive* iff we have $x = a/b$ for some positive integers a and b. It is said to be *negative* iff we have $x = -y$ for some positive rational y (i.e. $x = (-a)/b$ for some positive integers a and b.)

Definition 4.2.7 (Trichotomy of rationals)**.** Let x be a rational number. Then exactly one of the following three statements is true: (a) x is equal to 0. (b) x is a positive rational number. (c) x is a negative rational number/

Definition 4.2.8 (Ordering of the rationals)**.** Let x and y be rational numbers. We say that $x > y$ iff $x - y$ is a positive rational number, and $x < y$ iff $x - y$ is a negative rational number. We write $x \geq y$ iff either $x > y$ or $x = y$, and similarly define $x \leq y$.

Proposition 4.2.9 (Basic properties of order on the rationals)**.**

(a) (Order trichotomy) Exactly one of the three statements $x = y$, $x < y$, or $x > y$ is true.

(b) (Order is anti-symmetric) One has $x < y$ if and only if $y > x$.

(c) (Order is transitive) If $x < y$ and $y < z$, then $x < z$.

(d) (Addition preserves order) If $x < y$, then $x + z < y + z$.

(e) (Positive multiplication preserves order) If $x < y$ and z is positive, then $xz < yz$.